The Marginal Yield of LLM-Guided Alpha Search

Search Effort, Information and Investment Value

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Abstract

Language models can propose stock-return factors without limit, so the useful question is what an additional proposal is worth to an investor. We follow the gain from continued search through three stages: the training score the search optimizes, the accuracy of a joint return forecast, and the net return of a long-only portfolio. Paired experiments vary the search policy, the information available to the search and the downstream learner. The gain is stage-dependent. Training scores improve quickly and then saturate; price-volume search raises its best score by 17% within twelve further proposals. Little of this carries through. Repairing diagnosed formula defects raises factor scores while lowering forecast accuracy, and no policy changes net return by an amount distinguishable from zero. Tracing the libraries through the pipeline shows where the gains thin out: new factors overlap more with features the model already uses than the factors they replace, re-estimation offsets part of their contribution, and the stocks they add earn about what the displaced stocks earned, so transaction costs are not the cause. Research effort on factor search is best judged by the return it delivers, not by the score it optimizes.

Keywords: alpha discovery; large language models; search budget; return prediction; portfolio construction; information sets

1 Introduction

How much of the gain from alpha search reaches the portfolio? A research team can discover a better-scoring formula without improving the portfolio that uses it. The formula must first add information to a model that combines it with existing signals. Any change in that model's forecast must then move holdings enough to matter after trading costs. We measure this transmission from search progress to investment value as research effort accumulates.

Large language models make the question practical. They generate readable formulas, revise existing signals and combine economic reasoning with numerical feedback (Cheng and Tang, 2024; Wang et al., 2025). In such a setting research effort can buy several different things: more proposals, more refinement of existing proposals, or access to a broader information set. Reporting the best formula from each run compresses these choices into one number. Investors, however, act on a fitted model and a trading rule, and the response of each to additional search has to be measured on its own.

We call the change in a measured outcome from additional candidate proposals the *marginal yield of search*. A *factor library* is the set of selected formulas supplied to a predictive model, and the *joint predictor* combines that library with a common set of base features. The *training frontier* is the best training score among all eligible candidates seen so far. We distinguish it from the quality of the selected library, from the

accuracy of the joint forecast and from portfolio return. Two designs supply the evidence. In *matched-library continuation*, four search policies start from identical libraries and receive the same number of proposals. In *discovery from scratch*, the library starts empty and the search receives either price-volume fields or an expanded dictionary of valuation, flow and financial-statement fields. Two controls then separate the value of information from the value of searching for transformations of it: a linear model given sixteen raw fields directly, and a comparison in which both predictors already receive the same raw fields.

The question arose from a long archived search. Its best eligible training score rose by 2.6% over the first thirteen iterations and did not move again, while the remaining iterations consumed a third of all recorded model output and kept replacing library members. A flat frontier, a changing library and a moving portfolio can follow different paths; the controlled comparisons measure how far they do. Appendix E documents that search.

Across these designs, better training scores do not carry through to better forecasts or returns in any systematic way. Repair of diagnosed formula defects raises factor training scores as much as any policy, yet lowers the joint forecast's accuracy, and its portfolio return is equivalent to that of a mixed-operator policy within half a percentage point a year. An expanded information set finds a strong candidate almost immediately and changes which formulas are kept, yet a regularized linear model given sixteen training-selected raw fields is not outperformed by the searched libraries at equal dimension and needs no generation calls. Price-volume search meanwhile lifts its frontier by 17% during continuation while its net return does not reliably change. When both predictors already receive the raw fields, formulas composed from the expanded dictionary keep a small, significant forecasting advantage, and the return difference is bounded within one percentage point a year. Following the libraries through the pipeline shows where the gains thin out: entering factors overlap more with the existing features than the factors they replace, re-estimating the common coefficients reduces their contribution in point estimate, and a gradient-boosted learner changes three to four times as much holding weight without a detectable change in return, because the stocks it adds earn about the same as those it removes; fees contribute almost nothing to any of these changes.

This paper contributes a measurement: the marginal yield of search reported as a sequence of stage-wise increments under a common predictor and account, with information supply, initialization cost and model dimension controlled. The practical question it answers is which next investment, another proposal, better information or a better selection objective, is most likely to improve the final portfolio.

2 Search effort, information and investment value

Formulaic alphas are compact expressions of price, volume and accounting data whose cross-sectional ranks forecast returns (Kakushadze, 2016). Their discovery has been automated by evolutionary and learning-based search (Zhang et al., 2020; Cui et al., 2021) and by reinforcement learning that optimizes a collection whose members work jointly (Yu et al., 2023). Language models predict returns from text (Lopez-Lira and Tang, 2023), extend the factor search space to readable economic hypotheses (Cheng and Tang, 2024; Wang et al., 2025), and in agentic systems jointly propose factors and models (Li et al., 2025); the broader program-search literature pairs generative proposals with numerical evaluation (Romera-Paredes et al., 2024). Recent systems structure the exploration further: AlphaAgent regularizes proposal novelty and complexity (Tang et al., 2025), QuantaAlpha evolves complete mining trajectories (Han et al., 2026), and AlphaMemo records edits and their outcomes to guide later proposals under a fixed budget (Yu et al., 2026). These systems are evaluated on final factor quality or on discovery efficiency under a fixed budget. Our object is the path by which additional proposals change an incumbent solution, measured after a common predictive model and portfolio rule.

The relevant incumbent depends on the economic task. A search procedure may retain the formula with the highest training score, select a diverse library, or maximize a joint predictor's objective, and these

are different optimization problems. Machine-learning predictors absorb many characteristics at once (Gu et al., 2020; Chen et al., 2024; Leippold et al., 2022; Kelly and Xiu, 2023), so a regularized predictor can reallocate weight across the common features when the library changes and partially absorb new information. A portfolio applies a further transformation, such as holding the top predicted decile at fixed dates. Score differences near the selection threshold move holdings; movements among stocks that stay selected do not trade.

This connects formula discovery to the economics of characteristics and trading costs. The investment value of an additional characteristic depends on its effect on an implementable portfolio net of costs (DeMiguel et al., 2020; Novy-Marx and Velikov, 2016), and a large archive of candidate predictors raises the familiar selection problem (Harvey et al., 2016; Harvey and Liu, 2015; Bailey et al., 2014), which backtesting protocols for machine-learning strategies address by fixing the evaluation before results are seen (Arnott et al., 2019). We therefore compare libraries under common prediction and account rules, evaluate every scheduled budget rather than a selected best, and report paired uncertainty for the contrasts that carry the conclusions.

Information breadth is a separate input. If a search with broader data outperforms a price-volume search, the difference may come from the raw fields, from their nonlinear composition, or from a larger effective feature set. Two controls address these alternatives. The first selects sixteen raw fields on training data and fits the same Ridge model used for learned formulas, matching model dimension. The second gives all added raw fields to both predictive models before comparing the searched additions, holding direct information supply fixed.

3 Research design

Every experiment shares one measurement pipeline: candidate proposals pass through the library selector into a sixteen-member factor library, the library is joined to Alpha158 in a Ridge predictor, and the predictor drives a top-decile portfolio. Table 1 summarizes what each comparison asks, varies and holds fixed. All three share the formula grammar, the library selector, the Ridge predictor and the portfolio rule described in Section 3.5.

3.1 Budgets and outcomes

An *opportunity* is one scheduled candidate proposal. Invalid expressions, duplicates and failed proposals consume their opportunities, so the budget counts attempted research rather than successful output. Let q denote cumulative opportunities and L_q the selected library at that budget. Library capacity is sixteen.

Let $\mathcal{T}$ be the set of training decision dates, f_t the signed cross-section of a factor on date t, $r_{t\to t+10}$ the ten-session next-open return and corr_S the Spearman correlation. The *training IC* of a factor, the training frontier F_q and the library median M_q are

$$\mathrm{IC}(f) = \frac{1}{|\mathcal{T}|}\sum_{t\in\mathcal{T}} \mathrm{corr}_S\big(f_t,\, r_{t\to t+10}\big), \qquad F_q = \max_{f\in\mathcal{C}_q} \mathrm{IC}(f), \qquad M_q = \mathrm{median}_{f\in L_q}\,\mathrm{IC}(f), \tag{1}$$

where $\mathcal{C}_q$ is the set of eligible candidates seen by budget q. The frontier is nondecreasing by construction, whereas the library median can move in either direction because the selector weighs turnover and redundancy as well as IC: L_q consists of the first sixteen admissible candidates in nondominated order on the pair $(\mathrm{IC}(f),\, -\mathrm{TO}(f))$, with $\mathrm{TO}(f)$ the factor's rank turnover, subject to $\max_{g\in L_q,\, g\neq f} |\bar\rho(f,g)| < 0.995$ for the mean signed daily correlation $\bar\rho$ between percentile signals (Appendix A). Prediction quality, called RankIC, is

$$\rho_q = \frac{1}{|\mathcal{T}'|}\sum_{t\in\mathcal{T}'} \mathrm{corr}_S\big(\hat y_{q,t},\, r_{t\to t+10}\big), \tag{2}$$

Table 1. The three controlled comparisons. Each varies one research input while the predictor and account are common.

Comparison	Research question	What varies	What is held fixed
Matched-library continuation (Section 3.2)	Does the search policy change what eight further proposals deliver at the training, prediction and portfolio stages?	Proposal policy: mixed-operator, repair-directed, independent generation, genetic	Starting library and history (48 states, eight replicates), budget of eight opportunities, grammar, selector, 2010–2019 fit, 2020–2022 evaluation
Discovery from scratch (Section 3.3)	What do information supply and initialization add, and how does searched transformation compare with direct use of raw fields at equal dimension?	Input dictionary: price-volume or 227 added fields; a direct-field Ridge control with sixteen training-selected raw fields	Empty learned-factor start, 24 initialization plus twelve continuation opportunities, four paired repetitions, sixteen library members, 174 model columns, 2020–2022 evaluation
Common raw inputs (Section 3.4)	Once both predictors already receive the added raw fields, does composing them inside formulas add forecast or portfolio value?	Fields the search may compose: price-volume only, or also 25 added fields	Both predictors receive the 25 raw fields, 24 paired starting libraries, eight opportunities, 2010–2015 fit, evaluation in 2016–2019 and 2020–2022 without refitting

where $\hat{y}_{q,t}$ is the fitted model's prediction and $\mathcal{T}'$ the evaluation decision dates. Investment value R_q is the annualized arithmetic net return of the portfolio, and return differences are reported in percentage points (pp) per year. For an additional h opportunities the marginal change in an outcome $Y \in \{F, M, \rho, R\}$ is the increment $Y_{q+h} - Y_q$. A positive frontier increment means a better candidate was observed; the other three increments measure whether the library, the joint forecast and the portfolio also improved. Opportunity counts give a common budget axis; observed token consumption gives a separate resource axis.

3.2 Continuation from matched libraries

Starting libraries of sixteen members are sampled from higher- and lower-training-quality portions of a historical candidate pool. For each sampled pair, deterministic preparation supplies states after zero, sixteen and 48 prior proposals, so that one *initialization replicate* yields six related starting states. Each state's library, candidate history and training feedback are copied to four continuation policies:

- **Mixed-operator search** cycles through semantic mutation, crossover, defect repair and independent generation.
- **Repair-directed search** uses recorded defects to choose revisions, reserves every fourth opportunity for independent generation, and alternates mutation and crossover when no eligible defect exists.
- **Independent generation** proposes new formulas without editing parent expressions.
- **Genetic search** alternates parameter mutation and compatible subtree crossover under deterministic rules and makes no language-model calls.

The policies share a formula grammar, training selector, library capacity, feedback schedule and numerical evaluator. The three language-model policies use Claude Opus 5.5 with separate contexts after the common start. Every proposal, successful or not, counts towards the budget.

The primary comparison uses eight initialization replicates: 48 starting states and 192 continuation paths, each compared with its own starting library after eight opportunities. Within a replicate the six states are averaged before replicates receive equal weight. An earlier cohort of four replicates (24 states) supplies complete intermediate budget curves for the mixed-operator and repair-directed policies.

3.3 Discovery from scratch

The information experiment starts with empty learned-factor libraries, empty formula histories and no inherited candidate pool. Four paired search repetitions compare two input dictionaries. The *price-volume* dictionary contains open, high, low, close, volume, traded amount and the volume-weighted average price. The *expanded* dictionary adds all 227 locally available fields that meet fixed training coverage and variation criteria, covering valuation, capitalization, turnover, trading flows and financial statements.

Each path first receives 24 initialization opportunities in six batches of four proposals. Initialization uses the dictionary and grammar without numerical feedback; earlier formulas are shown only to avoid duplicates. Twelve continuation opportunities follow in six two-proposal batches, cycling twice through mutation, crossover and independent generation with statistics for the selected library supplied. Outcomes are evaluated at total budgets of zero, eight, sixteen, 24, 26, 28, 30, 32, 34 and 36 opportunities. Every path reaches sixteen members by the end of initialization; earlier libraries with fewer members use zero columns for unfilled positions. The downstream Alpha158 base, Qlib's standard set of 158 price-volume features (Yang et al., 2020), is common to every model.

The *direct-field Ridge* control screens the 227 added fields by training-only absolute IC, requires at least 70% coverage, greedily removes near-clones at an absolute correlation of 0.995, and supplies the sixteen selected raw fields to the same preprocessing and Ridge specification as the learned libraries. Every final predictor therefore has 174 columns: 158 common features plus sixteen additions. The control learns coefficients on existing fields and requires no generation calls. The two search arms have equal opportunity counts, but their token costs differ because field descriptions and feedback differ in length; all initialization and continuation calls enter the resource totals.

3.4 Formula composition with common raw inputs

A third comparison isolates the value of composing added fields when both predictive models already receive those fields. Both arms share Alpha158 and 25 eligible raw fields: nine daily valuation, turnover, size or share-count fields, thirteen financial-statement fields, two trading-flow fields and one share-unlock pressure field. With a sixteen-member learned library each predictor has 199 columns.

Twelve replicate groups each provide a higher- and a lower-quality starting library, giving 24 paired starting libraries rebuilt from historical first-generation proposals with 2010–2015 training statistics. Both arms receive eight opportunities per starting library. Price-volume search composes price-volume inputs only; expanded-information search may also compose the 25 added fields. Prompts show the current formulas with their training IC, coverage and turnover, and the selector applies the same individual-factor criteria to both arms. Search, feature scaling and model fitting use 2010–2015, and the fixed models are evaluated in 2016–2019 and 2020–2022 without refitting.

3.5 Market data, predictor and portfolio

The equity panel is drawn from the China Securities Index 500 (CSI500), the mid-capitalization segment of the Chinese A-share market, using dated constituent masks and common stock-date support within each comparison. The continuation and from-scratch studies fit on 2010–2019 and evaluate on 2020–2022; the common-input study uses the earlier split described above. Prices share one adjusted basis across arms. In the from-scratch study, daily added fields become available from the following market session. In the common-input study, daily valuation and trading-flow fields enter at the close with trades at the next open, and financial records enter after their recorded announcement date or a documented historical-date proxy.

The predictive base is Alpha158. Features are transformed to average-tie cross-sectional percentiles on finite dated-member support, missing values are imputed with training means, and columns are centred and scaled with training statistics; degenerate columns become zero. Ridge uses a fixed penalty of one and an unpenalized intercept,

$$\min_{a,\beta} \frac{1}{n}\|y - a\mathbf{1} - X\beta\|^2 + \|\beta\|^2. \tag{3}$$

Columns, scaling parameters and coefficients are fixed before evaluation, and the common features' coefficients are re-estimated whenever the library changes.

A signal formed after session t predicts the ten-session next-open return $y_{i,t} = O_{i,t+11}/O_{i,t+1} - 1$, where O is the adjusted open; training labels that cross the fitting boundary are excluded. Every ten sessions the portfolio targets equal capital in the top predicted decile and executes at the next open. Accounts are long-only, start from cash, use fractional units and pay two basis points per side. Positions without an execution quote stay locked at their last price, and there is no forced terminal liquidation. The 2020–2022 evaluation has 728 valuation dates, 727 daily returns and 72 mature prediction dates; the two-period common-input evaluation has 168 mature prediction dates and 1,697 daily returns.

3.6 Paired inference and economic margins

Search repetitions share market dates. For the matched-state continuation comparison, paired effects are first averaged over the six states of each initialization replicate and Student-t intervals are then computed across the eight replicate effects, so that initialization is the inference unit. The four-replicate and from-scratch comparisons use their four paired repetition effects.

Three continuation questions were specified in advance: return equivalence between repair-directed and mixed-operator search within ± 0.5 pp per year, tested with two one-sided tests (Schuirmann, 1987) and a 90% interval; noninferiority of genetic search to each language-model policy at the same margin, with Holm adjustment (Holm, 1979) across the three comparisons; and a joint directional pattern of higher member training IC, lower model RankIC and higher return for repair-directed search relative to its starting library. All other intervals are 95%.

For discovery from scratch the primary contrast is expanded minus price-volume net return at 36 opportunities. Repetition intervals describe variation across the four paired runs; separately, 2,000 paired circular block draws of market dates (Politis and Romano, 1992) with block lengths of twenty and sixty sessions describe calendar variation for the fixed runs. The direct-field model is a single fitted comparator, so contrasts with it condition on that model.

For the common-input comparison, the registered analysis uses both the 24 paired starting-library effects (Student-t, 23 degrees of freedom) and calendar uncertainty from heteroskedasticity and autocorrelation consistent (HAC) standard errors (Newey and West, 1987) with one lag for decision-date RankIC and five lags for daily returns. The registered sensitivity rule requires the pooled RankIC effect to keep its sign under HAC lags zero through six, in both evaluation periods and in at least nine of twelve initialization groups. The larger of the two p-values enters Holm adjustment across the two primary endpoints, and the pooled

return contrast is compared with a prespecified ± 1 pp economic band. The two types of uncertainty are reported side by side rather than merged into one enlarged sample.

4 Results

4.1 Better factor training scores can lower joint prediction

All four continuation policies raise the median training IC of the library in the eight-replicate comparison (Table 2), while their investment outcomes sit in a narrow band. Starting libraries earn 14.312% a year net; the four policies range from 14.081% to 14.407%.

Repair-directed search gives the sharpest evidence of divergence across stages (Figure 1). Relative to the starting library its median member IC rises by 5.95×10^{-3} (95% interval $[2.81, 9.10] \times 10^{-3}$), its model RankIC falls by 0.60×10^{-3} ($[-0.95, -0.26] \times 10^{-3}$), and its annual return changes by -0.163 pp ($[-0.514, 0.189]$). Independent generation and genetic search raise both training IC and RankIC, and no policy reliably changes return. A stronger individual training score is therefore not a reliable guide to the direction of joint predictive change.

Repair-directed and mixed-operator search satisfy the prespecified economic equivalence criterion: their return difference is -0.115 pp with 90% interval $[-0.384, 0.154]$ and equivalence $p = 0.015$. Reallocating proposals between these two policies therefore has a small economic effect. Genetic search, which makes no model calls, differs from the mixed-operator, repair-directed and independent-generation policies by -0.183, -0.068 and -0.326 pp; its noninferiority at the 0.5 pp margin is not established (adjusted $p = 0.295$). The three-part directional hypothesis, which also required a positive return effect for repair, has joint $p = 0.845$: the training and prediction components hold, and the return component is centred slightly below zero; Appendix C gives the component tests. The earlier four-replicate cohort shows the same training–prediction divergence with a repair return increment of $+0.389$ pp against -0.163 pp here, so the divergence repeats across cohorts while the sign of the return increment does not (Appendix D).

4.2 More information changes the early frontier

The information dictionary changes discovery from the first proposals (Figure 2). By eight opportunities, expanded-information search reaches a mean best training IC of 0.0723, close to its final 0.0724. Price-volume search starts at 0.0603 and ends at 0.0763. The richer dictionary finds a strong incumbent early; the price-volume frontier keeps improving late. The early expanded-information frontier is the same candidate in all four repetitions, the negative twenty-session mean of the free-float turnover rate, proposed with or without a rank or z-score wrapper. It is a direct transformation of one added field rather than a composition of several.

Table 2. Four search policies after eight additional opportunities. Each entry averages six starting states within each of eight initialization replicates. Net return is annualized arithmetic after 2 bp per side; excess is the annualized active return over the CSI500 price index and IR its information ratio; turnover is two-way traded notional per rebalance in NAV units. The starting library is the shared zero-search reference.

Library	Median member IC	Model RankIC	Net return (%)	Excess (pp)	IR	Max DD (%)	Turnover
Starting library	0.01474	0.05930	14.312	8.17	0.73	−31.7	1.66
Mixed-operator	0.01802	0.05924	14.264	8.12	0.72	−31.8	1.66
Repair-directed	0.02069	0.05870	14.149	8.00	0.71	−31.9	1.65
Independent generation	0.02083	0.05948	14.407	8.26	0.74	−31.8	1.66
Genetic	0.01939	0.05979	14.081	7.93	0.71	−31.7	1.66

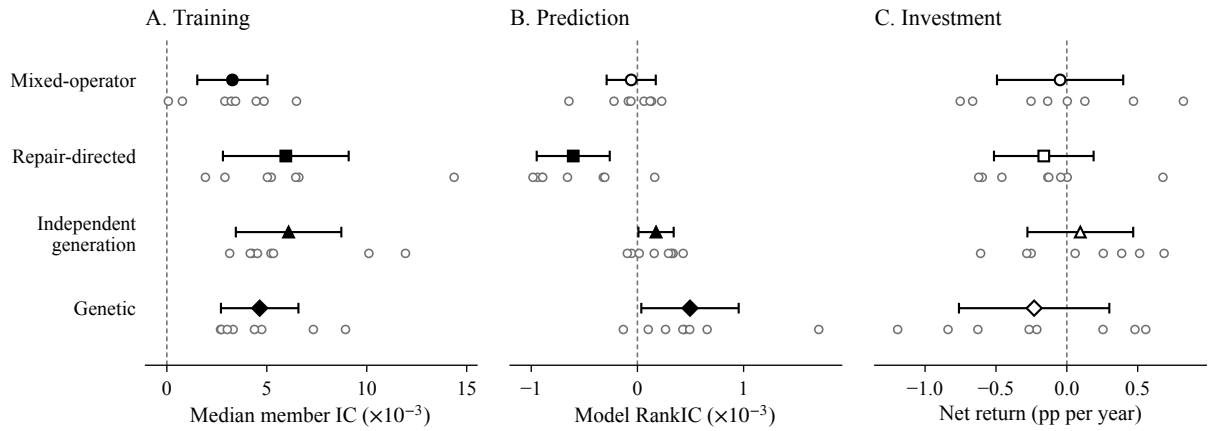


Figure 1. Repair-directed search raises factor training scores and lowers joint prediction, and no policy reliably changes return. Changes from the starting library after eight opportunities; open grey circles are the eight initialization replicates, markers and whiskers their mean and 95% Student-t interval, filled when the interval excludes zero.

The selected library follows a different pattern. At the end of initialization the median member IC is 0.0474 for price-volume and 0.0342 for expanded-information search, and after twelve further opportunities 0.0541 and 0.0364. The lower expanded-information median reflects the selection rule, which balances IC against turnover and redundancy rather than admitting the best individual candidates.

Prediction also responds early. Expanded-information search reaches RankIC 0.0653 at sixteen opportunities against 0.0596 for price-volume search; at the final budget the values are 0.0629 and 0.0618, a paired difference of 1.10×10^{-3} with repetition interval $[-1.14, 3.35] \times 10^{-3}$.

4.3 Training gains and investment gains separate

During price-volume continuation the training frontier rises from 0.0653 to 0.0763, a gain of 16.9% in the mean that comes from three of the four repetitions (0%, 12.5%, 23.9% and 32.9%). Median member IC rises by 6.62×10^{-3} and model RankIC by 0.58×10^{-3}. Net return moves from 14.896% to 14.657%, a change of -0.239 pp with repetition interval $[-4.210, 3.732]$ and twenty-session market-block interval $[-2.254, 1.867]$. Expanded-information continuation moves the frontier by 0.12×10^{-3}, raises median member IC by 2.22×10^{-3}, lowers RankIC by 0.86×10^{-3} and raises return by 0.148 pp ($[-5.197, 5.493]$ across repetitions, $[-1.709, 2.207]$ across blocks). One arm records a substantial training gain; neither records a reliable economic increment.

At the final budget the expanded-minus-price-volume return difference is -0.465 pp, with repetition

Table 3. Information, initialization and equal-dimension controls. Search rows average four repetitions; initialization ends at 24 opportunities and continuation at 36. Every row except the base uses Alpha158 plus sixteen additional columns. Excess and IR are against the CSI500 price index; turnover is 1.65 to 1.67 in every row.

Information and model	Stage	RankIC	Net return (%)	Excess (pp)	IR	Max DD (%)
Alpha158	Common base	0.05705	13.320	7.17	0.63	−31.9
Price-volume search	Initialization	0.06118	14.896	8.75	0.76	−32.2
Price-volume search	Final	0.06176	14.657	8.51	0.75	−32.1
Expanded-information search	Initialization	0.06372	14.045	7.90	0.68	−31.9
Expanded-information search	Final	0.06286	14.192	8.05	0.70	−31.6
Direct-field Ridge	Sixteen selected fields	0.06272	14.995	8.85	0.75	−32.8

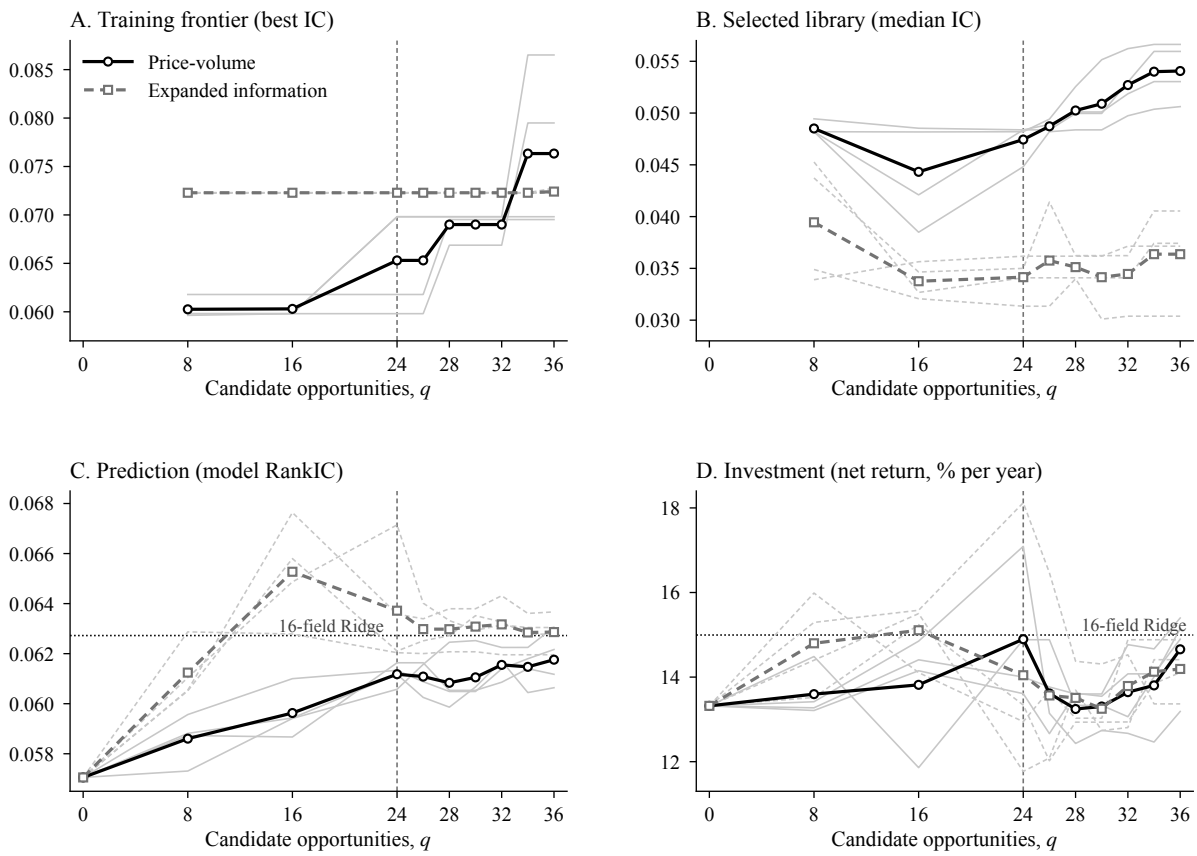


Figure 2. Expanded information lifts the training frontier and prediction early, while net return separates little between arms. Discovery from scratch over total opportunities q; thin lines are four repetitions, thick lines their means, the dashed vertical line the end of initialization, and the dotted reference the fixed sixteen-field Ridge model. Training IC is undefined at $q = 0$. The four expanded-information repetitions nearly coincide in panel A because they reach the same best candidate by $q = 8$; one repetition improves slightly at the final budget.

interval $[-2.374, 1.445]$ and market-block intervals $[-3.439, 2.528]$ and $[-3.384, 2.420]$ for twenty and sixty sessions (Table 12): the two information sets did not reliably differ in return. The marginal return to search is also non-monotone in point estimates. Over the final four opportunities, price-volume and expanded-information returns rise by 1.013 and 0.403 pp, neither reliably different from zero, and in the four-replicate continuation curves the gains over opportunities four to eight exceed those over zero to four while no curvature contrast is rejected. The evidence separates training progress from economic progress; it does not establish a curve of diminishing returns.

4.4 Direct-field weighting is a competitive reference

The direct-field control reaches RankIC 0.0627 and net return 14.995% with sixteen training-selected fields, essentially the expanded-information search mean of 0.0629 (Table 3). The search-minus-control return differences are -0.338 pp for price-volume search and -0.802 pp for expanded information, with repetition intervals $[-1.930, 1.255]$ and $[-1.815, 0.210]$ (Figure 3). The control raises return by 1.675 pp over Alpha158, with twenty-session market-block interval $[-2.630, 5.816]$. Because the comparison holds the final number of model columns fixed, it is a low-cost reference for what searched transformations add beyond the fields themselves.

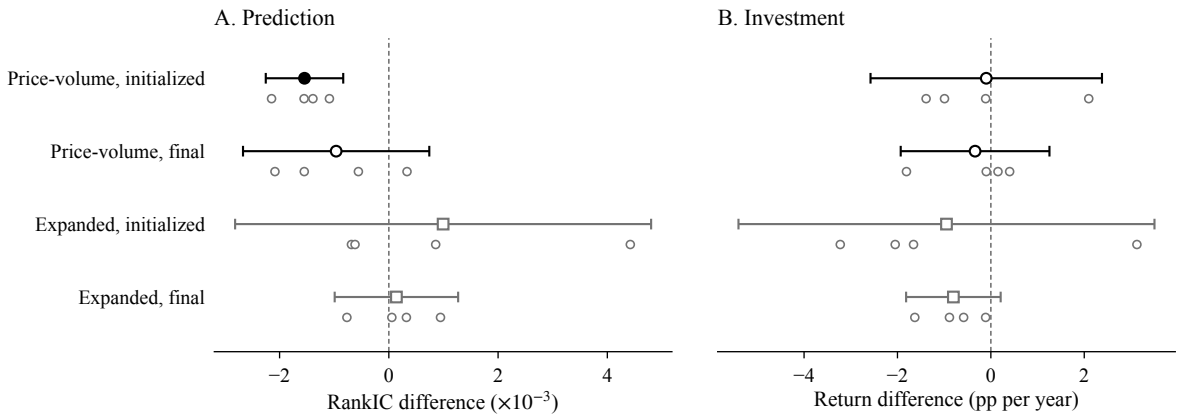


Figure 3. Searched libraries do not outperform a sixteen-field Ridge model of equal dimension. Differences are search minus control at the initialized (24) and final (36) budgets; open grey circles are four repetitions, markers and whiskers their mean and 95% interval conditional on the fixed control. Black marks price-volume and grey expanded information; a filled marker means the interval excludes zero.

The two pipelines select different expressions of the same information. Fourteen of the control's sixteen fields are trading flows and two are turnover measures. In the expanded-search final libraries, 52 of 64 member positions use added fields covering 23 distinct fields in six source families, and none uses a trading-flow field; during initialization, 59 of 92 parseable expanded-information formulas use added fields. The language model therefore draws heavily on the expanded dictionary, and its outcome reflects which information and transformations it chooses together with the common model's response.

Initialization is a material part of the resource requirement (Table 9). Across four paths per arm, price-volume search uses 175,910 observed tokens and expanded-information search 502,307, a ratio of 2.86 at equal opportunity counts, with initialization accounting for 74,032 and 235,744 tokens. The direct-field control screens the 227 fields numerically and uses no generation calls. Equal dimension, equal opportunity count and equal resource expenditure are therefore three distinct comparisons.

4.5 Composition keeps a forecasting edge beyond direct information supply

With both models given Alpha158 and the same 25 raw fields, the common-input comparison measures what composing those fields inside formulas adds. Across 24 paired starting libraries and the pooled 2016–2022 evaluation, expanded-information search has a RankIC advantage of 1.00×10^{-3} over price-volume search, with calendar-HAC interval $[0.23, 1.76] \times 10^{-3}$, starting-library interval $[0.66, 1.33] \times 10^{-3}$ and adjusted $p = 0.021$. The within-arm changes explain the advantage: price-volume search lowers RankIC by 1.15×10^{-3} from its starting library, whereas expanded-information search changes it by -0.16×10^{-3}. Composition with the expanded dictionary preserves most of the starting predictor's quality while price-volume revisions lose predictive content. The contrast is positive in both evaluation periods (Table 4).

The pooled return difference is -0.303 pp with calendar-HAC interval $[-0.810, 0.205]$, starting-library interval $[-0.580, -0.026]$ and adjusted primary $p = 0.242$. Both intervals lie inside the prespecified ± 1 pp band (Figure 4). Information therefore matters through composition even after its raw fields are supplied to the predictor, and the portfolio consequence of that forecasting edge is economically bounded under the top-decile rule.

Table 4. Formula composition under common raw information. All models use the same 183 base columns plus sixteen learned factors, are fit on 2010–2015 and evaluated without refitting. Entries average 24 paired starting libraries; returns are annualized arithmetic net percentages. Excess return, information ratio, drawdown and calendar-year returns for 2020–2022 are in Appendix B.

	2016–2019		2020–2022	
Library	RankIC	Net return (%)	RankIC	Net return (%)
Starting library	0.07311	7.647	0.06380	19.422
Price-volume final	0.07181	8.004	0.06285	19.895
Expanded-information final	0.07285	7.687	0.06378	19.612

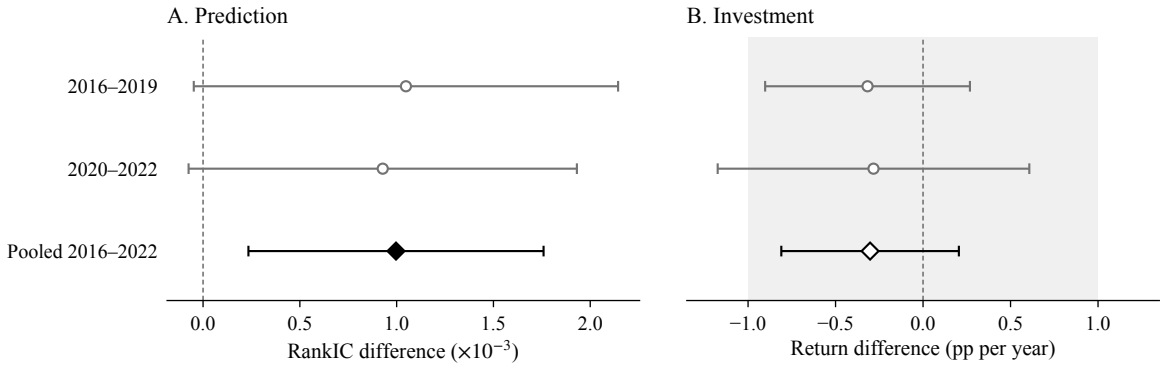


Figure 4. With common raw inputs, expanded-information composition keeps a significant forecasting advantage and a return difference inside the ± 1 pp band. Expanded minus price-volume search; whiskers are 95% calendar-HAC intervals, the pooled row is the registered primary comparison, and the shaded band is the prespecified tolerance. Grey rows are single periods and black the pooled estimate; a filled marker means the interval excludes zero.

5 Where the transmission weakens

The divergence between training scores, joint forecasts and returns raises the question of which link in the pipeline absorbs the search gains. This section follows the original continuation cohort, 24 starting states under four policies after eight opportunities, through each stage of the pipeline without any new proposal: members are replaced, the model is refit, holdings change and costs are charged. Four diagnostics isolate in turn the redundancy of entering factors with the existing features, the effect of refitting the common coefficients, the change in portfolio membership, and the parts of the return change that come from gross profit and from trading costs. A fifth refits the same libraries with a gradient-boosted learner, and a sixth follows both learners along the budget and decomposes the realized return into the stocks swapped, the weights on common holdings and the fees. All six use the 24 starting and 96 terminal libraries of the original cohort evaluated in Appendix D; that cohort is used because its saved fits are complete for all four policies. In this cohort the Ridge return gains are positive in point estimate, and only the repair-directed gain has a replicate interval that excludes zero; in the eight-replicate comparison of Section 4 they are not. The diagnostics therefore ask why member-score gains translate so weakly, not why returns fall.

5.1 Members, predictions and holdings

Table 6 follows each policy from member replacement to the account. Eight opportunities replace 4.7 to 5.2 of the sixteen members and raise median training IC by 3.4 to 5.3 thousandths, yet the refitted Ridge model's predictions keep a rank correlation of 0.992 to 0.994 with the starting model's, the top decile overlaps by

92.0 to 93.5%, and the two accounts' holdings differ by 6.5 to 8.0% of weight. The fee drag changes by less than 0.003 pp per year, so the net gain of 0.19 to 0.58 pp in point estimate is profit and loss earned on the 7% of holdings that differ, not a cost effect. Descriptive rank correlations within replicates, after removing state means, show the training gain negatively related to the prediction gain (-0.30 to -0.40), the prediction change tracking the holding change (0.92 to 0.95), and the holding change only weakly related to the profit gain (0.13 to 0.30). The chain weakens at its first link, from member quality to joint prediction, and again at its last, from holdings to profit.

What the search changes in the portfolio can be written as an identity. For two equal-weight baskets of the same size on the same date, the change in the basket's ten-session return equals the fraction of names replaced times the return difference between the stocks added and the stocks removed; averaged over dates with the replaced fraction as weight, the identity holds for the pooled figures as well. Table 5 applies it to the first and second four opportunities under both learners. Ridge libraries replace 4.6% and then 5.3% of top-decile names per date, and the added stocks outperform the removed ones by 4.7 and 25.6 basis points per ten sessions. LightGBM replaces 22.8% and 24.0%, about twelve names per date against three, but its added stocks outperform the removed ones by 6.9 and 0.4 basis points. Every spread has a calendar interval that includes zero (Figure 5). An exact daily decomposition of the realized account return over the second four opportunities attributes the Ridge change of $+0.316$ pp to $+0.979$ pp from the stocks added, -0.656 pp from the stocks removed, -0.007 pp from weight differences on common holdings and -0.001 pp from fees; the LightGBM terms are $+4.644$, -4.647, -0.003 and -0.001 pp, summing to -0.007 pp. Fees therefore play no part in the weak transmission: the learner that replaces four times as many names replaces them with stocks that earn about the same as those they displace. The full cross-section and the invested tail can also move apart. Under LightGBM over the second four opportunities, genetic search raises model RankIC by 0.53 thousandths while top-decile ten-session return falls by 2.6 basis points and the account by 0.62 pp, whereas independent generation raises all three.

5.2 Entering factors overlap more with the base than the factors they replace

A member-level decomposition of the 120 saved Ridge fits describes each factor by its maximum absolute correlation with a single Alpha158 feature, by the R^2 of its projection on all 158 base features, and by its absolute standardized coefficient (Table 7). Entering members overlap more with the base than the members they replace under every policy, by 0.005 to 0.12 in projection R^2. Under repair-directed search, whose entrants have training IC 0.0249 against 0.0062 for the leavers, entrants receive larger coefficients than leavers; under the other three policies they receive 12 to 17% smaller ones. The change in the fitted training prediction has four components, the entering members, the removed members and the re-weighting of the retained members and of the base features, and they largely offset one another: the standard deviation of the

Table 5. The marginal return is earned on the stocks swapped, not lost to fees. Original cohort, four policies averaged over 24 starting states. Replaced is the share of top-decile names changed per date between the two budgets, with the mean number of names; spread is the ten-session return of the added minus the removed stocks in basis points, weighted by the replaced share; the remaining columns are the exact decomposition of the change in annualized net return into stocks added, stocks removed, weight differences on common holdings and fees, in percentage points per year.

Learner	Opportunities	Replaced (%)	Names	Spread (bp)	Added	Removed	Common	Fees	Net
Ridge	First four	4.6	2.3	4.7	$+0.786$	-0.714	-0.004	-0.000	$+0.068$
Ridge	Second four	5.3	2.7	25.6	$+0.979$	-0.656	-0.007	-0.001	$+0.316$
LightGBM	First four	22.8	11.3	6.9	$+4.787$	-4.406	-0.013	$+0.000$	$+0.369$
LightGBM	Second four	24.0	12.0	0.4	$+4.644$	-4.647	-0.003	-0.001	-0.007

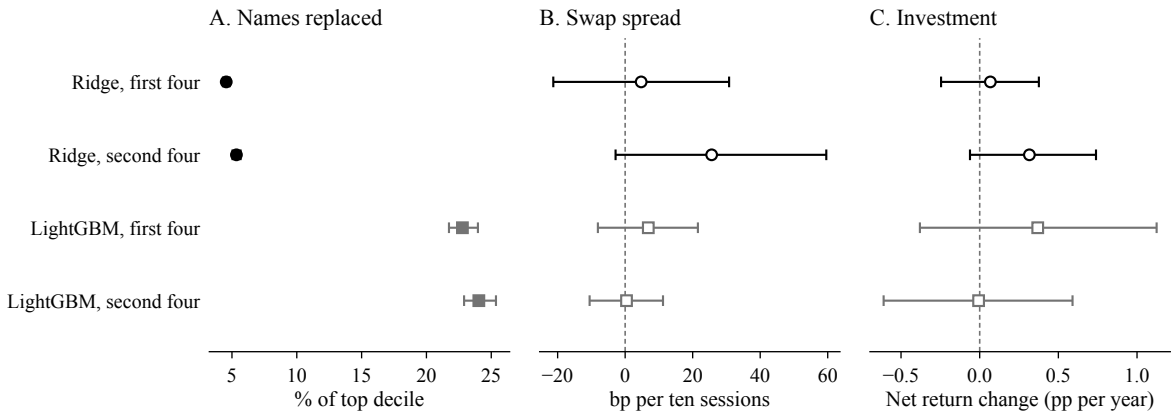


Figure 5. LightGBM replaces four times as many names as Ridge for no better ones. Original cohort, four policies averaged; rows are the first and second four opportunities under each learner; the spread is the ten-session return of the stocks added minus those removed; whiskers are 95% twenty-session calendar intervals, filled when they exclude zero.

Table 6. The same 96 library changes followed through Ridge and LightGBM. Four-replicate means over 24 starting states after eight opportunities. New members are replaced members of sixteen; rank correlation and top-decile overlap compare terminal with starting predictions; holdings changed is the mean absolute weight difference between the two accounts; gains are terminal minus starting, RankIC in thousandths and net return in percentage points per year.

Learner	Policy	New members	Rank corr.	Overlap (%)	Holdings changed (%)	RankIC gain	Net gain
Ridge	Mixed-operator	5.17	0.992	92.0	8.0	-0.23	$+0.58$
Ridge	Repair-directed	5.00	0.994	93.1	6.9	-0.69	$+0.39$
Ridge	Independent generation	4.67	0.993	92.5	7.5	-0.23	$+0.19$
Ridge	Genetic	4.92	0.994	93.5	6.5	$+0.01$	$+0.37$
LightGBM	Mixed-operator	5.17	0.895	72.9	27.2	$+0.37$	$+0.59$
LightGBM	Repair-directed	5.00	0.904	74.3	25.9	-0.16	$+0.31$
LightGBM	Independent generation	4.67	0.902	73.7	26.4	$+0.92$	$+0.59$
LightGBM	Genetic	4.92	0.905	74.3	25.9	$+1.02$	-0.04

net change is 31 to 38% of the sum of the components' standard deviations. Coefficients shrink entrants only modestly, and under repair-directed search not at all, so a library whose members score better individually need not predict better jointly. Within replicates, the entrants' projection R^2 is negatively related to the prediction gain (descriptive rank correlations -0.10 to -0.53).

5.3 Refitting the common coefficients

Because the common features' coefficients are re-estimated whenever the library changes, the net gain mixes the new members' own contribution with a reallocation of weight among the old features. Holding the Alpha158 coefficients at their starting values while the library component changes raises net return by 0.60 pp per year (four-replicate interval $[0.15, 1.04]$); allowing the base coefficients to re-adjust, which is the actual procedure, gives 0.38 pp ($[-0.08, 0.84]$). Re-estimating the base coefficients therefore lowers the gain by 0.21 pp in point estimate; the interval excludes zero only for sixty-session blocks ($[-0.10, 0.53]$ across replicates, $[-0.04, 0.46]$ and $[0.02, 0.40]$ across twenty- and sixty-session blocks). New base coefficients with the old library move return by -0.11 pp ($[-0.42, 0.19]$), and a Shapley split attributes 0.55 pp ($[0.12, 0.98]$) to the library and -0.16 pp ($[-0.42, 0.09]$) to the base re-weighting. Frozen coefficients change model Ran-

Table 7. Entering factors overlap more with the base than exiting factors and are shrunk only modestly for it. Means over 120 saved Ridge fits of the original cohort; $|\rho|_{\max}$ is the maximum absolute correlation with one Alpha158 feature, R^2 the projection on all base features, $|\beta|$ the absolute standardized coefficient in units of 10^{-4}, and net/sum the standard deviation of the net change in the fitted training prediction as a percentage of the sum of its four components' standard deviations.

	Entering			Exiting			
Policy	$\lvert\rho\rvert_{\max}$	R^2	$\lvert\beta\rvert$	$\lvert\rho\rvert_{\max}$	R^2	$\lvert\beta\rvert$	Net/sum (%)
Mixed-operator	0.363	0.309	2.73	0.348	0.304	3.10	35.1
Repair-directed	0.454	0.381	3.48	0.354	0.320	2.97	32.3
Independent generation	0.476	0.414	2.57	0.331	0.293	3.09	37.6
Genetic	0.354	0.314	2.57	0.340	0.299	2.97	30.5

kIC by -0.25 thousandths ($[-0.51, 0.01]$). The refit may absorb part of what the new members add, and the part that survives does not reliably raise the return either.

5.4 The portfolio rule and trading costs

Table 8 applies six account rules to the same 96 sets of predictions. Under the primary rule, long the top decile rebalanced every ten sessions, net returns are 14.4 to 14.8% and the search gains 0.19 to 0.58 pp. Rebalancing the same predictions daily lowers the level to 7.1 to 7.6% and gives negative point estimates for the three language-model policies, with every interval including zero. Top-half long-only and long–short rules place every gain within ± 0.15 pp. The 36 policy contrasts and 30 rule-by-policy interactions all have Holm-adjusted p-values of one. Within the primary rule the change in fee drag is negligible, whereas across rules the level moves by seven points: the account rule, not the library, sets the cost side of the return. Weighting the top decile by predicted rank instead of equally changes the mean search gain from 0.38 to 0.33 pp (interaction -0.05, $[-0.31, 0.21]$).

5.5 The same libraries under a gradient-boosted learner

To test whether the weak transmission is a property of Ridge, the original cohort's 24 starting and 96 terminal libraries were re-evaluated under a second, untuned learner, LightGBM at the fixed specification of Appendix G. The nonlinear learner moves the second and third links of the chain (Table 6, Figure 6): rank correlation with the starting model falls to 0.895 to 0.905, top-decile overlap to 73 to 74%, and holdings changed rise to 25.9 to 27.2%, which is 18.9 to 19.4 pp of weight more than under Ridge with every policy's interval excluding zero (Holm $p \leq 0.0007$). The return does not follow. LightGBM net gains are $+0.59$, $+0.31$, $+0.59$ and -0.04 pp with every interval including zero, every learner-by-policy interaction on return

Table 8. Net-return gain from the starting library under six account rules, in percentage points per year, 24 states per policy at 2 bp per side. Rules hold the top 10% or top 50% of predicted scores long, or the top and bottom 50% long and short with half the capital on each side, rebalanced every ten sessions or daily on the same predictions.

Rule	Mixed-operator	Repair-directed	Independent generation	Genetic
Long top 10%, ten-session (primary)	$+0.58$	$+0.39$	$+0.19$	$+0.37$
Long top 10%, daily	-0.12	-0.16	-0.43	$+0.05$
Long top 50%, ten-session	$+0.07$	$+0.08$	-0.05	$+0.08$
Long top 50%, daily	-0.05	-0.08	-0.14	-0.09
Long–short 50%, ten-session	$+0.07$	$+0.06$	-0.07	$+0.08$
Long–short 50%, daily	-0.05	-0.07	-0.14	-0.10

has Holm-adjusted $p = 1$, and the repair-minus-mixed contrast is -0.27 pp ($[-1.54, 1.05]$ by twenty-session blocks) against -0.19 pp under Ridge. At this specification LightGBM also predicts less well than Ridge in this panel (starting-library RankIC 0.031 against 0.059, arithmetic annual net return 10.2% against 14.2%), and its terminal libraries earn 10.2 to 10.8% against 11.4% for Alpha158 alone, differences that include zero. Along the budget the two learners respond at different times (Figure 7). Averaged over the four policies, the first four opportunities add 0.07 pp under Ridge and 0.37 pp under LightGBM, the next four 0.32 and -0.01 pp, so the endpoint gains of 0.38 and 0.36 pp arrive early under one learner and late under the other, and no policy shows a diminishing pattern under both (second-four gains of $+0.49$, $+0.27$, $+0.15$ and $+0.36$ pp under Ridge against $+0.04$, $+0.18$, $+0.37$ and -0.62 pp under LightGBM). LightGBM moves 16 to 20% of holding weight after only two opportunities, against 2.7 to 3.4% under Ridge, while its RankIC stays near 0.032 at every budget; the calendar intervals of the pooled segment gains include zero. A learner that changes three to four times as much holding weight therefore produces no detectable change in return from the same library changes: under this specification the learner changed how much of the search reached the holdings, not, detectably, what the holdings earned. The re-evaluation covers this cohort only; the eight-replicate, from-scratch and common-input experiments have been run under Ridge.

6 Implications for alpha research

6.1 A search frontier is one part of a research frontier

The best observed training score answers whether search has found a better candidate under its own objective. It is an extreme-value statistic over the candidate archive, and it can rise while the selected library becomes less useful, the joint forecast weakens or the portfolio stands still. The from-scratch price-volume arm shows this directly: a 16.9% frontier gain accompanies a -0.24 pp return change. The eight-replicate continuation adds directional evidence at the next stage, with higher member training IC and lower joint RankIC under repair-directed search. The joint model rewards information that stays useful alongside existing inputs, and the portfolio responds only to rankings that cross its selection threshold. Gains can therefore accumulate at the search stage and be absorbed by model reweighting or by the portfolio rule, and the appropriate next research step depends on where the transmission weakens.

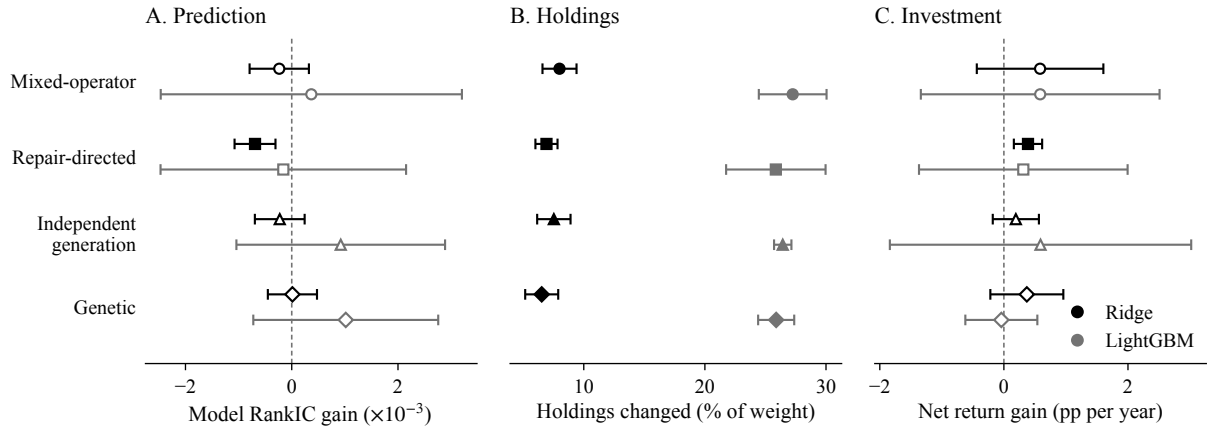


Figure 6. A learner that changes three to four times as much holding weight shows no detectable change in return. The same 96 library changes refit with Ridge (black) and LightGBM (grey); markers and whiskers are four-replicate means and 95% intervals, filled when the interval excludes zero.

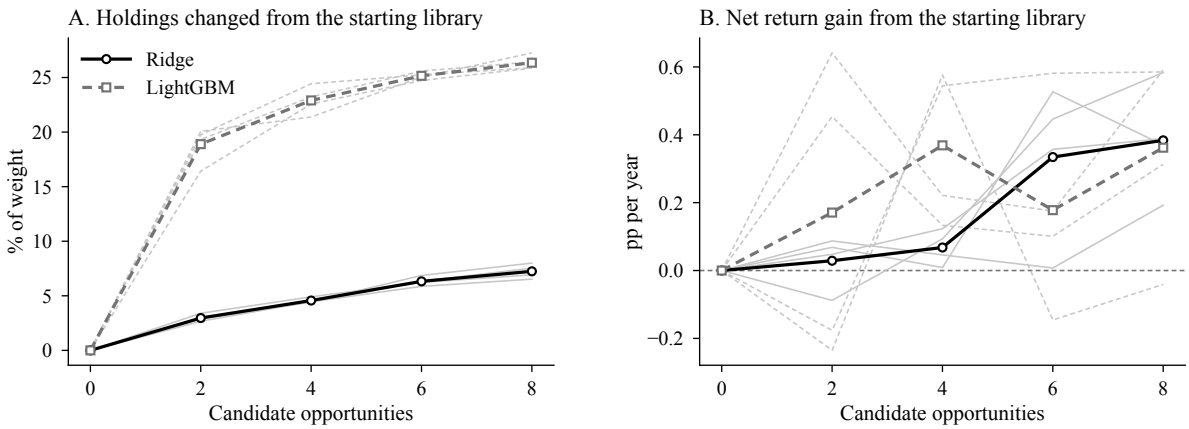


Figure 7. Under matched libraries, Ridge and LightGBM gain similar amounts by eight opportunities but at different points of the budget. Thick lines are equal-weight averages over the four policies, thin lines the policies; Ridge black, LightGBM grey dashed.

6.2 Information acquisition and formula refinement are different investments

Expanded information changes early discovery and the formulas kept in the library, and the from-scratch design shows that this begins during initialization, before any performance feedback. Its cost begins there too: initializing the richer dictionary accounts for almost half of that arm's tokens. The sixteen-field control uses the same information pool differently, ranking raw fields on training data and letting Ridge estimate their joint weights, and its outcome puts the question on the incremental value of transformations once useful fields are available; an equal-capacity control with random formulas points the same way (Appendix F). In the common-input experiment, composition keeps a small forecasting advantage with a bounded portfolio difference. A team deciding where to spend its next unit of effort can thus compare another formula proposal with direct use of an informative field, and the results suggest selecting formulas for the information they add to the existing predictor rather than for their individual training score.

6.3 Investment value depends on prediction and portfolio rules

The top-decile equal-weight account uses predictions only on rebalance dates and only through membership in the selected group, so changes within the group do not trade, and Section 5 shows that the group changes by about 7% of weight under Ridge and 26% under LightGBM for the same libraries. Joint estimation also reallocates weight among the common features as the library changes and reduces the new members' contribution in point estimate. Among the three studies, the equivalence of repair-directed and mixed-operator search within half a point and the bounded composition effect within one point are the sharper economic statements; the from-scratch return intervals are several points wide.

6.4 Limitations

The controlled comparisons use one market, one primary portfolio rule and Ridge as the primary learner; the LightGBM re-evaluation covers the original continuation cohort at one fixed specification. The 2020–2022 window served as the development period of the research programme and lies inside the language model's training corpus, so absolute return levels inherit both caveats while the paired contrasts share the exposure. The from-scratch and continuation studies rest on four and eight paired repetitions, with repetition and market-date uncertainty reported separately. Financial-statement fields are a current-vintage vendor

snapshot rather than a revision-free historical vintage.

7 Conclusion

Additional factor-search effort can move the training frontier substantially without a matching change in investment value. Across matched continuation, discovery from scratch and common-input composition, the same pattern recurs: candidate quality, joint prediction and portfolio return respond to different objectives. Repair-directed search improves individual factor scores while weakening joint prediction. Expanded information reshapes early discovery, searched libraries do not outperform a sixteen-field Ridge control at equal dimension, entering factors overlap more with the base than those they replace, a nonlinear learner moves far more holding weight without a detectable change in return because its swaps are no better, fees play no part, and composition with the expanded dictionary keeps a significant forecasting edge whose portfolio effect is bounded. For quantitative research the marginal yield of search is therefore best reported as a sequence of measured increments, and search is best judged on the outcome that is paid for.

Data and code availability. Companion materials contain factor formulas, input-field definitions, search protocols, saved predictions, account summaries, statistical specifications and figure-generation code, together with an index linking each reported result to its source table. Licensed raw vendor data are excluded.

Disclosure. Claude Opus 5.5 generated the factor proposals that are the object of the experiments. Large language models also assisted with numerical diagnostics, reference verification and editing of the manuscript. All results and all text were checked by the author, who takes responsibility for the analysis and the article.

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A Selection and information controls

The library selector requires a preassigned formula sign, at least 70% average dated-member coverage and at least one valid training decision date. A valid score date has at least twenty labelled stocks, at least two labelled stocks in the preformed top decile and nonconstant ranked cross-sections. The selector orders admissible candidates by nondominated sorting on $(\mathrm{IC}(f), -\mathrm{TO}(f))$, breaking ties within a front by crowding distance and then deterministically, and admits candidates in that order while $\max_{g \in L} |\bar{\rho}(f, g)| < 0.995$ against the members already admitted, where $\bar{\rho}$ is the mean signed daily correlation between percentile signals. Dominance tolerance is 10^{-12} and there is no positive-IC floor. Up to sixteen factors are retained with different trade-offs between score, turnover and redundancy; the joint predictor's incremental performance is not an admission criterion.

For the from-scratch study the expanded dictionary admits every available field with at least 70% average training coverage and nonconstant values; no field-count cap is imposed. Financial records keep their recorded units and date-alignment rules, and rank transformations precede cross-unit operations where the grammar requires. The direct-field control screens this same dictionary and selects sixteen fields by absolute individual training IC subject to the coverage and near-clone rules; the selected fields receive the same cross-sectional percentile transformation and standardization as learned formulas, and all selection precedes the evaluation period.

The from-scratch experiment contains eight paths, 288 scheduled opportunities and 96 generation calls. There are 274 numerical training measurements; seven statically invalid proposals, one duplicate and six opportunities lost to response-format failures complete the count. Of the measured proposals, 254 satisfy admission and twenty fail coverage; each final library has sixteen members.

Table 9. Generation resources for discovery from scratch. Totals cover four paths per information arm and include input, cached and output tokens. Numerical field screening and model fitting are additional computation.

Information	Initialization tokens	Continuation tokens	Total
Price-volume	74,032	101,878	175,910
Expanded	235,744	266,563	502,307
Total	309,776	368,441	678,217

The common-input composition experiment contains 48 paths and 384 candidate opportunities in 192 proposer calls.

B Account statistics by calendar year

Table 10 reports the journal-convention account statistics for every row of Tables 2, 3 and 4 on the 2020–2022 development period, computed from the saved daily equity paths. Excess return and the information ratio are against the CSI500 price index, whose own calendar-year returns close the table. The index is a price index without dividends, so excess returns are overstated by the dividend yield.

C Primary replication and uncertainty

Table 11 gives the eight-replicate continuation tests, each replicate contributing the mean of its six starting states. The training and prediction components of the directional hypothesis hold in their prespecified directions; the return component determines the joint result.

Table 10. Account statistics, 2020–2022. Net return and excess are annualized arithmetic percentages and percentage points; IR and Sharpe are annualized at a zero risk-free rate; max DD is the maximum drawdown of the equity path in percent; turnover is two-way traded notional per rebalance in NAV units; calendar-year columns are compounded returns in percent.

Row	Net	Excess	IR	Sharpe	Max DD	Turnover	2020	2021	2022
Continuation, eight replicates, eight opportunities									
Starting library	14.31	8.17	0.73	0.63	−31.7	1.66	35.6	19.4	−13.4
Mixed-operator	14.26	8.12	0.72	0.62	−31.8	1.66	35.9	19.3	−13.6
Repair-directed	14.15	8.00	0.71	0.62	−31.9	1.65	35.2	19.1	−13.4
Independent generation	14.41	8.26	0.74	0.63	−31.8	1.66	36.8	19.1	−13.8
Genetic	14.08	7.93	0.71	0.62	−31.7	1.66	35.4	19.2	−13.7
Discovery from scratch									
Alpha158	13.32	7.17	0.63	0.58	−31.9	1.67	35.1	16.6	−13.5
Price-volume search, Initialization	14.90	8.75	0.76	0.65	−32.2	1.66	38.5	16.3	−11.5
Price-volume search, Final	14.66	8.51	0.75	0.64	−32.1	1.66	39.4	16.8	−13.0
Expanded-information search, Initialization	14.04	7.90	0.68	0.62	−31.9	1.65	34.2	18.3	−12.3
Expanded-information search, Final	14.19	8.05	0.70	0.63	−31.6	1.66	36.0	18.7	−13.3
Direct-field Ridge	14.99	8.85	0.75	0.68	−32.8	1.66	39.8	17.0	−12.2
Common raw inputs, 2020–2022 period									
Starting library	19.42	13.28	1.16	0.85	−30.4	1.65	46.7	17.7	−5.9
Price-volume final	19.89	13.75	1.20	0.86	−30.5	1.65	46.5	18.6	−5.4
Expanded-information final	19.61	13.47	1.18	0.86	−30.0	1.65	47.0	17.9	−5.8
CSI500 price index	6.15	—	—	—	—	—	18.4	16.5	−19.7

Table 11. Prespecified continuation contrasts. Return units are percentage points per year. The equivalence interval is 90%; all other intervals are 95%. Component p-values are one-sided in their prespecified directions.

Contrast	Estimate	Interval	p
Repair minus mixed: return	-0.1149	$[-0.3843, 0.1545]$	0.0151
Repair minus start: member IC ($\times 10^{-3}$)	5.95	$[2.81, 9.10]$	0.0014
Repair minus start: RankIC ($\times 10^{-3}$)	-0.60	$[-0.95, -0.26]$	0.0021
Repair minus start: return	-0.1625	$[-0.5140, 0.1890]$	0.8448

The genetic noninferiority comparisons use the same half-point margin against each language-model policy with Holm adjustment across the three; their joint adjusted p-value is 0.295. The joint training-up, prediction-down, return-up test for repair-directed search has $p = 0.8448$.

Table 12. Return contrasts from discovery from scratch, in percentage points per year. Search intervals use four paired repetitions; market intervals resample paired daily returns in twenty-session blocks, conditional on those repetitions.

Contrast	Difference	Search 95% interval	Market 95% interval
Expanded minus price-volume, final	-0.465	$[-2.374, 1.445]$	$[-3.439, 2.528]$
Price-volume, final minus initialized	-0.239	$[-4.210, 3.732]$	$[-2.254, 1.867]$
Expanded, final minus initialized	0.148	$[-5.197, 5.493]$	$[-1.709, 2.207]$
Expanded final minus direct-field Ridge	-0.802	$[-1.815, 0.210]$	$[-5.434, 4.021]$
Direct-field Ridge minus Alpha158	1.675	—	$[-2.630, 5.816]$

For expanded minus price-volume at the final budget, sixty-session blocks give $[-3.384, 2.420]$ pp. The final RankIC difference is 1.10×10^{-3} with repetition interval $[-1.14, 3.35] \times 10^{-3}$ and two-sided $p = 0.216$. All final-budget comparisons use the scheduled endpoint; no best budget is selected.

In the common-input comparison the two primary pooled endpoints take the larger of the starting-library and calendar p-values before Holm adjustment, giving adjusted $p = 0.02078$ for RankIC and $p = 0.24235$ for return. The pooled RankIC effect is positive under HAC lags zero through six and in eleven of twelve initialization groups. Omitting the single most influential prediction date raises the lag-zero p-value from 0.0151 to 0.0252, just above the registered 0.025 threshold; the primary estimate uses all dates. Period-specific intervals are shown in Figure 4.

D Original cohort and longer-budget curves

All four policies have eight-opportunity evaluations at each of the original 24 starting states (Table 13). Intermediate budgets for independent generation and genetic search cover four of those states.

Table 13. Original 24-state comparison at eight opportunities. Entries average six states within each of four initialization replicates and then the four replicate means. Returns are annual arithmetic net percentages.

Library	Median member IC	Model RankIC	Net return (%)
Starting library	0.01446	0.05876	14.209
Mixed-operator	0.01787	0.05852	14.793
Repair-directed	0.01977	0.05806	14.598
Independent generation	0.01936	0.05853	14.402
Genetic	0.01845	0.05877	14.579

The original cohort supplies complete intermediate budgets for the mixed-operator and repair-directed policies (Table 14). The net-return gain over the first four opportunities is 0.093 pp for mixed-operator and 0.123 pp for repair-directed search; over the next four it is 0.490 and 0.265 pp. Seven exploratory return-curvature contrasts produce no Holm-adjusted rejection at the 5% level.

Table 14. Original continuation curves. Annual net returns are percentages; paired differences and intervals are percentage points per year across four initialization replicates.

Opportunities	Mixed	Repair	Repair minus mixed	95% interval
0	14.209	14.209	0.000	—
2	14.121	14.257	0.136	$[-0.799, 1.071]$
4	14.303	14.332	0.030	$[-0.932, 0.992]$
6	14.656	14.566	-0.089	$[-1.483, 1.304]$
8	14.793	14.598	-0.195	$[-1.035, 0.645]$

E Historical search: frontier growth and simple controls

E.1 The long-search setting

The historical study follows a search through 21 iterations, beginning with 240 proposals and adding 96 opportunities per iteration, with 32 factor places allocated across return prediction, weak-stock exclusion, portfolio complementarity and style implementation. The analyses below use the eighteen-member prediction library. Inputs are adjusted daily price, volume and traded amount, and the forecast target is the ten-session next-open return of the controlled studies.

The historical selection score is a partial rank association. On each date, factor ranks and return ranks are separately projected on an intercept and ranked price-volume controls, and their residual correlation

is averaged within training blocks and then equally across 2010–2012, 2013–2015, 2016–2017 and 2018–2019. The controls are log one plus mean traded amount over ten sessions, the twenty-session return, the 230-session return lagged twenty sessions, and squared five-session returns summed at lags zero, five, ten and fifteen. Eligibility requires a positive score in each block and sufficient coverage. This objective differs from the ordinary training IC of the controlled comparisons. Parent selection used training statistics, and development feedback at two intermediate checkpoints supplied cluster and task labels. The initial, training-peak and final libraries are taken at iterations zero, thirteen and 21, the peak being the first attainment of the trajectory's maximum.

E.2 A stationary frontier with continuing research effort

The best eligible score rises from 0.044659 to 0.045824, a 2.61% increase, and stays there through the final iteration (Figure 8). Cumulative recorded model output reaches 4.291, 14.640 and 21.685 million tokens at the initial, peak and final libraries, so search after the peak uses a further 7.046 million output tokens across all four search tasks, including thinking output and failed sessions.

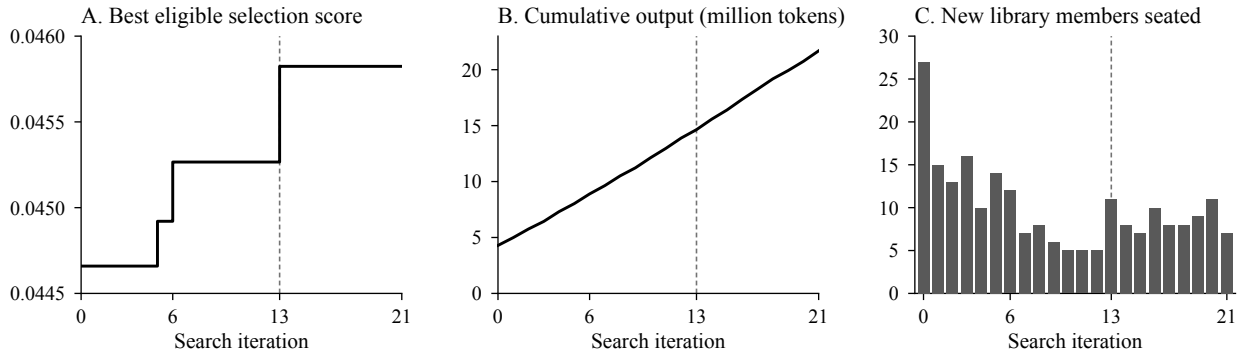


Figure 8. The historical frontier stopped rising at iteration thirteen while model output and library renewal continued. The dashed line marks the first attainment of the maximum selection score. Panel B is cumulative recorded output across all four search tasks; panel C counts newly generated candidates seated in any of the 32 library places at each iteration, where iteration zero is the 240-proposal initialization and each later iteration adds 96 proposals.

Library renewal continues after the frontier flattens. The last seven iterations retain sixty newly generated offspring across tasks, and newborn retentions per million recorded output tokens are 16.27, 8.08 and 9.82 in the successive seven-iteration periods. Retention efficiency therefore varies nonmonotonically while the best score is flat, which is why the library and the downstream portfolio are measured alongside the best candidate.

E.3 Offspring relative to parents and simple combinations

Table 15 measures each offspring's training-score increment against its strongest declared parent or, for two-parent operations, their equal-weight rank blend, using the same partial-correlation score. Pair availability is reported because some proposals lack a scored control, and the operation mix reflects the adaptive search policy.

Among proposals with a scored control, semantic mutation improves on it in 25.2% of cases, crossover in 17.4% and crossover followed by mutation in 12.0%. Against the equal-weight blend alone, the two crossover operations improve in 88 of 322 and 51 of 266 pairs. A common-stock diagnostic for 32 groups changes one increment's sign and leaves the strongest-control identity unchanged.

Table 15. Historical offspring versus the strongest available parent or equal-weight rank blend. Mean increments are differences in the historical selection score.

Operation	Proposals	Paired	Positive pairs (rate)	Positive/ proposals	Mean increment
Deterministic mutation	323	254	76 (29.9%)	23.5%	−0.003757
Semantic mutation	246	242	61 (25.2%)	24.8%	−0.010261
Crossover	323	322	56 (17.4%)	17.3%	−0.012850
Crossover then mutation	267	266	32 (12.0%)	12.0%	−0.013048

Positive/proposals counts confirmed successes among all allocated opportunities; unpaired proposals have unknown increments.

F Random and searched formulas at equal nominal capacity

A separate control starts from the four original higher-quality libraries before preparation, each with sixteen factors, and compares the starting library with the addition of eight random formulas and with the addition of all eight mixed-operator proposals (Table 16). Each augmented Ridge model has 182 nominal columns including the 158 base features; the starting model has 174. Random formulas use the same operator families and a fixed legal draw distribution with depths one to four, without conditioning on any score; one draw is made per library.

Table 16. Equal-addition control at four original starting libraries. Means are annualized arithmetic net returns after 2 bp per-side fees; paired 95% t intervals have three degrees of freedom and refer to 2020–2022.

Library or paired contrast	Mean (% or pp/year)	95% interval (pp/year)
Starting sixteen members	13.954	—
Starting plus eight random expressions	14.714	—
Starting plus eight mixed-operator proposals	14.125	—
Random minus start	+0.760	$[-2.623, +4.143]$
Search minus start	+0.171	$[-2.576, +2.918]$
Search minus random	−0.589	$[-4.733, +3.555]$

Search exceeds the random additions in two of the four paired comparisons, with mean difference -0.589 pp per year and interval $[-4.733, 3.555]$. Random additions supply seven usable columns in two libraries and eight in the other two; searched additions supply eight in each, and degenerate columns keep zero values after scaling. The control holds nominal feature capacity and grammar fixed and complements the direct raw-field comparison by contrasting two ways of creating transformed features.

G Historical investment performance across evaluation choices

G.1 Models, libraries and uncertainty

The historical initial, training-peak and final libraries each contain eighteen factors. LightGBM is fit to the Alpha158 base, to each library alone, and to each library added to the base; Ridge comparisons use the base alone and the peak and final augmentations. Training covers 2010–2019. LightGBM uses 300 trees, learning rate 0.05, 31 leaves, minimum leaf size 200 and an L_2 penalty of one; Ridge uses the main specification. Parameters are common across libraries within each learner.

Portfolios hold the equally weighted top predicted decile, rebalance every ten sessions and trade at the next open. The original fee is two basis points per side. A higher-cost variant charges five basis points on purchases and fifteen on sales, and a dated-tax variant charges five basis points commission per side plus sale

stamp duty of ten basis points before 28 August 2023 and five thereafter, with target quantities recomputed under each schedule.

This appendix reports compound annual growth rates (CAGR) over each account's calendar span, whereas the controlled experiments report annualized arithmetic net return. Paired 95% intervals resample the same daily log-return vectors across compared accounts with 2,000 circular block draws of twenty and sixty sessions, anchoring the initial entry cost once; they describe calendar variation conditional on the fixed libraries and models. Signal intervals resample two or six consecutive decision-date RankIC observations, with Newey–West intervals as a further check.

G.2 Development performance depends on the learner

All historical development accounts share the original stock-eligibility intersection and 728 valuation dates in 2020–2022 (Table 17). "Base + peak" and "Base + final" augment Alpha158 with the corresponding eighteen factors.

Table 17. Development accounts under the original eligibility rule, 2020–2022. CAGRs are percentages; augmentation differences are percentage points.

Learner	Features	RankIC	Net CAGR 2/2 bp	Net CAGR 5/15 bp	Added vs. Base 5/15 bp (pp)
LightGBM	Base	0.02926	8.345	4.984	–
LightGBM	Initial only	0.01849	2.916	-0.564	–
LightGBM	Peak only	0.01994	8.511	4.892	–
LightGBM	Final only	0.00219	6.303	2.741	–
LightGBM	Base + initial	0.02940	11.714	8.281	+3.297
LightGBM	Base + peak	0.02619	12.275	8.867	+3.884
LightGBM	Base + final	0.01842	11.725	8.269	+3.285
Ridge	Base	0.05705	10.864	7.368	–
Ridge	Base + peak	0.05926	11.178	7.693	+0.325
Ridge	Base + final	0.05818	13.949	10.393	+3.025

RankIC uses 72 mature rebalance dates; all accounts share 728 valuation dates, and fee variants reuse the fixed selections.

The final-minus-peak augmentation difference is -0.550 pp of CAGR under LightGBM, with twenty-session interval $[-6.790, 6.297]$, and $+2.771$ pp under Ridge, with interval $[-1.670, 7.363]$; the higher-cost convention gives -0.599 and $+2.701$. For standalone LightGBM libraries the final-minus-peak RankIC is -0.01775, with twenty- and sixty-session intervals $[-0.03351, -0.00158]$ and $[-0.03217, -0.00461]$. Library, predictor and portfolio rankings therefore need separate measurement.

G.3 Later-period results and trading costs

The later-period comparison applies the six training-fitted pipelines to 2023–2025 without refitting. Every arm uses dated constituent membership, a finite Alpha158 input and a positive execution price, with missing added features handled by the model's preprocessing. The period has 727 valuation dates and 73 rebalances (Table 18); Table 19 reports risk and calendar-year performance.

At the original fees the final-minus-peak CAGR is -0.052 pp under LightGBM, with twenty- and sixty-session intervals $[-5.921, 5.586]$ and $[-4.935, 5.028]$, and -1.129 pp under Ridge, with twenty-session interval $[-4.280, 1.767]$, although Ridge's RankIC rises from 0.01555 to 0.01804. The learner interaction in the late-library return difference is -1.077 pp with interval $[-7.257, 5.518]$. At the higher fixed fees the late differences are -0.023 and -1.094 pp for LightGBM and Ridge, and under dated tax -0.026 and -1.104 pp. Trading costs lower account levels materially, while similar turnover limits their effect on

Table 18. Historical library comparisons in 2023–2025 with training-fitted models. Net CAGRs are percentages; excess is dated-tax portfolio CAGR minus CSI500 price-index CAGR, in percentage points.

Learner	Features	RankIC	2/2 bp	5/15 bp	Dated tax	Excess (pp)
LightGBM	Base	0.00894	6.800	3.371	4.211	−4.248
LightGBM	Base + peak	0.01615	7.299	3.772	4.624	−3.835
LightGBM	Base + final	0.00944	7.247	3.749	4.597	−3.861
Ridge	Base	0.01342	13.071	9.458	10.342	+1.883
Ridge	Base + peak	0.01555	11.770	8.209	9.081	+0.622
Ridge	Base + final	0.01804	10.641	7.115	7.977	−0.482

CSI500 price-index CAGR is 8.459%. Dated tax is 5 bp commission per side plus sale stamp duty of 10 bp before 28 August 2023 and 5 bp thereafter.

the relative ranking. The excess column compares portfolio CAGR with the price index without dividend reinvestment in the benchmark.

Table 19. Risk and calendar-year performance in 2023–2025 under dated-tax costs. Drawdowns and annual returns are percentages.

Learner	Features	Sharpe	Max DD	Turnover	2023	2024	2025
LightGBM	Base	0.292	−31.92	42.56	−5.19	−2.61	22.52
LightGBM	Base + peak	0.306	−37.21	43.57	−6.38	−1.04	23.56
LightGBM	Base + final	0.305	−40.02	43.21	−6.80	−3.22	26.83
Ridge	Base	0.543	−27.60	42.27	−4.13	15.03	21.74
Ridge	Base + peak	0.492	−30.24	42.15	−5.00	11.74	22.19
Ridge	Base + final	0.448	−29.96	42.16	−2.89	6.10	22.10

Sharpe uses daily account returns, a zero risk-free rate and 252-session annualization. Turnover is the 2/2-bp account's annualized two-way traded notional divided by prior NAV. Annual returns use the opening-to-opening account convention.

G.4 Stock eligibility affects point rankings

The original development intersection requires at least one finite base feature and one finite feature in each of four inherited factor families. A matched-eligibility comparison removes the inherited-family filters and applies the later period's membership and base-feature rule to development, with models and dates unchanged (Table 20); executable stock-date observations rise from 35,683 to 36,419.

Table 20. Development final-minus-peak CAGR under the original and matched eligibility rules. Estimates and intervals are percentage points; all models are fixed.

Learner	Stock rule	2/2 bp	95% CI: 2/2 bp	5/15 bp
LightGBM	Original intersection	−0.550	[−6.790, +6.297]	−0.599
LightGBM	Matched eligibility	+0.320	[−5.982, +7.359]	+0.243
Ridge	Original intersection	+2.771	[−1.670, +7.363]	+2.701
Ridge	Matched eligibility	+2.219	[−1.960, +6.474]	+2.163

Intervals use paired twenty-session blocks. Matched eligibility removes four inherited feature-family filters and requires dated membership, a finite Alpha158 feature and a positive next-open execution price.

Under matched eligibility the LightGBM final-minus-peak CAGR becomes $+0.320$ pp with interval $[-5.982, 7.359]$, and the Ridge difference $+2.219$ pp with interval $[-1.960, 6.474]$. Changing the eligibility

rule can change a point ranking with identical models and dates, which is why the evaluated stock support belongs in any economic comparison.

H US selection from a fixed formula pool

A companion study evaluates formula reuse in a cohort of 503 current US share classes, using total-return-adjusted daily OHLCV data, 2010–2019 training and 2020–2022 evaluation. The sample is defined by current constituents.

The common base contains 157 OHLCV features, omitting the unavailable VWAP feature. The compared additions are ten compatible China training-peak formulas, nine compatible China final formulas, and sixteen formulas selected on US training data from a fixed historical pool of 113 compatible formulas, of which near-clone pruning removes two; training IC orders the remainder. Formulas and their directions are inherited, so no new US formulas are generated. Each feature set is fit with Ridge under the main account rules, and paired uncertainty uses 2,000 synchronized circular block draws (Table 21).

Table 21. US fixed-pool selection, current 503-share-class cohort, evaluation 2020–2022, Ridge fit on 2010–2019. Panel A: 2 bp per-side fees; Ann. is annualized arithmetic net return, CAGR compound growth, MDD maximum drawdown, all in percent, Sharpe at a zero risk-free rate. Panel B: equal-date RankIC and paired return differences from Base in pp per year with 95% synchronized block-bootstrap intervals.

Panel A: Portfolio outcomes

Inputs	Features	Ann.	CAGR	Sharpe	MDD
Base	157	22.731	14.953	0.541	−51.674
+ China training-peak	167	22.983	15.204	0.546	−53.487
+ China final	166	22.785	14.951	0.540	−52.535
+ US-selected	173	24.768	17.386	0.591	−52.026

Panel B: Prediction and uncertainty

Inputs	RankIC	Δ Ann.	20-session interval	60-session interval
Base	0.005235	—	—	—
+ China training-peak	0.003914	+0.252	$[-3.134, +3.088]$	$[-2.973, +3.297]$
+ China final	0.004737	+0.054	$[-2.201, +1.884]$	$[-2.119, +1.950]$
+ US-selected	0.005096	+2.036	$[-0.396, +4.402]$	$[-0.133, +4.323]$

The US-selected library's annual arithmetic return estimate is $+2.036$ pp above the base, with intervals $[-0.396, 4.402]$ and $[-0.133, 4.323]$ at the two block lengths that include zero, while its RankIC is 0.005096 against 0.005235 for the base and CAGR rises from 14.953% to 17.386%. The account improvement and the prediction ordering differ here as well. The return estimate stays close to two percentage points above the base at the higher transaction charges (Table 22), in a companion setting with its own information set, market period and current-constituent cohort.

Table 22. Current-cohort US sensitivities. Calendar-year returns and annualized arithmetic net returns at each per-side cost are percentages; turnover is mean two-way traded notional per rebalance in NAV units.

Inputs	2020	2021	2022	2 bp	5 bp	10 bp	Turnover
Base	43.514	40.248	−24.610	22.731	21.754	20.126	1.287
+ China peak	39.492	42.116	−22.955	22.983	22.007	20.381	1.286
+ China final	40.096	42.800	−24.154	22.785	21.813	20.192	1.281
+ US selected	47.944	43.522	−23.913	24.768	23.782	22.140	1.298